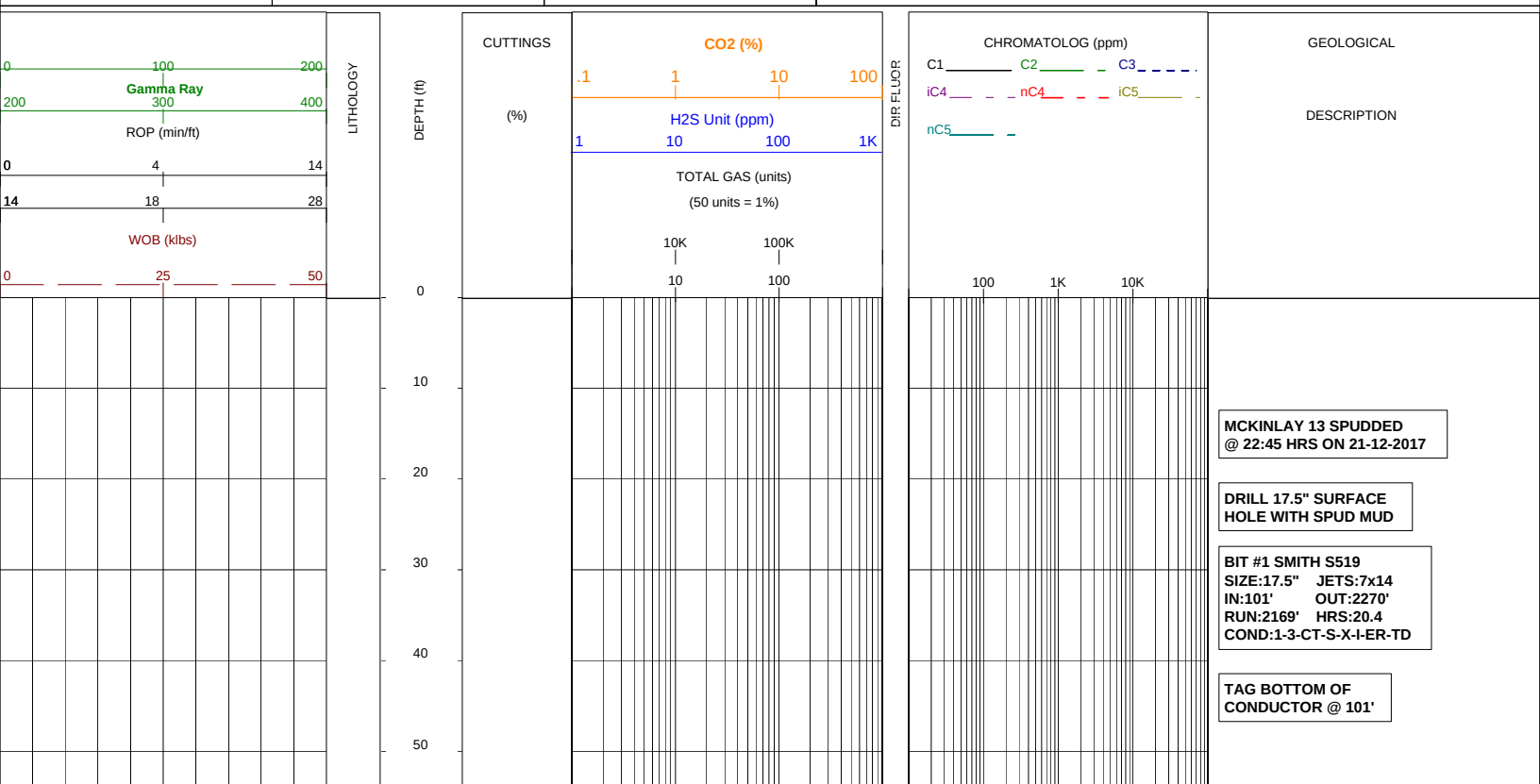
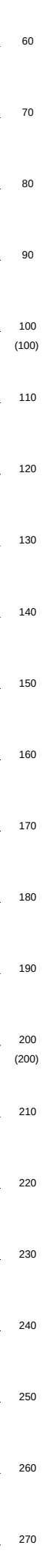
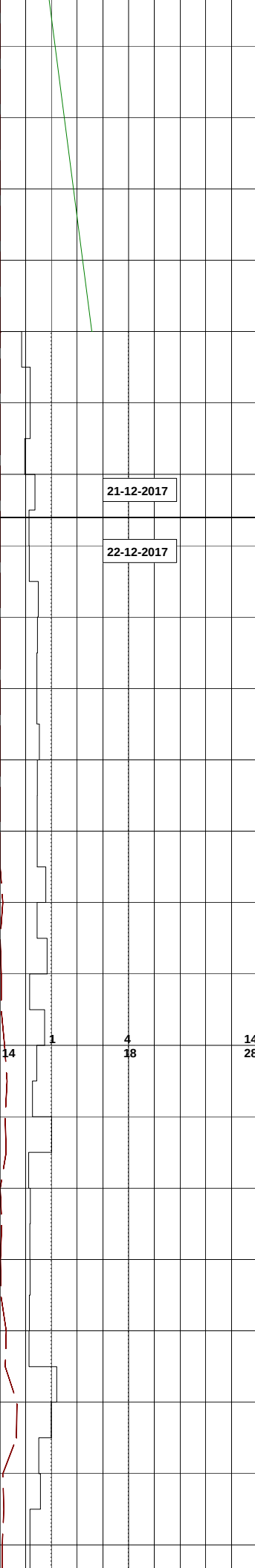


Field : MCKINLAY	Rotary Table : 149.8'	Rig : ENSIGN 971	Open Hole:	Cased Hole:	Loggers : SEAN KETTLE
Permit : PPL234	Ground Level : 126.3'	Spud Date: 21-12-2017	17.5" 2270'	13.375" 2260'	MATHEW TOWNSEND
State : SOUTH AUSTRALIA	GDA 94 Co-ordinates :	TD Date : 01-01-2018	12.25" 5077'	9.625" 5068'	VAHID JALEH
Country : AUSTRALIA	Lat. : 28°29'06.29"S	Total Depth : 10593'MDRT 4061'TVDRT	8.5" 10593'	5.5" LINER 10583'	
Scale : 1/240 5":100"	Long. : 140°28'38.63"E	Final Status : C&S OIL WELL			

LITHOLOGY	ACCESSORIES	DRILLING DATA	ABBREVIATIONS
 Coarse sandstone  Medium sandstone  Fine sandstone  Cement  Siltstone  Carb Siltstone  Metamorphic rock  Claystone  Limestone  Coal-lignite	 Pyrite  Siderite  Glauconite  Iron  Mica  Chert  Calcareous  Dolomitic  Carbonaceous  Fossils	 Casing shoe  Bit Trip  Wiper Trip  Core  Dst  Survey MUD DATA MW - Mud Weight (lb/gal) FV - Funnel Viscosity (s/qt) PV - Plastic Viscosity (cps) YP - Yield Point (lb/100ftsq) Gel - Gel Strength (10sec) WL - Water Loss (cc/30min) pH - Acidity / Alkalinity Ck - Cake (32nd/inch) Sol - Solids (% vol) Cl - Chlorides (mg/l)	ABBREVIATIONS BOPD - Barrels of Oil Per Day BWPD - Barrels of Water Per Day CG - Connection Gas CO - Circulate Out COND - Condensate c/c - Crush Cut DST - Drill Stem Test FLOW - Flow Rate (gal/min) GCM - Gas Cut Mud GCW - Gas Cut Water GTS - Gas To Surface INJ - Injection of Mist (bbls/hr) LCM - Lost Circulation Material MMCFD - Million Cubic Feet / Day NGTS - No Gas To Surface NOTS - No Oil To Surface NR - No Returns OCM - Oil Cut Mud OG - Over Gauge OH - Open Hole OTS - Oil To Surface Q - Flow Rate REC - Recovery Rmf - Resistivity mud filtrate ROP - Rate Of Penetration RPM - Revolutions Per Minute RTSTM - Rate Too Small To Measure Rw - Resistivity water r/r - Ring Residue SCFM - Standard Cubic Ft/Min (air) SGCM - Slightly Gas Cut Mud SPM - Strokes Per Minute SPP - Stand Pipe Pressure SWC - Side-Wall Core TG - Trip Gas WOB - Weight On Bit





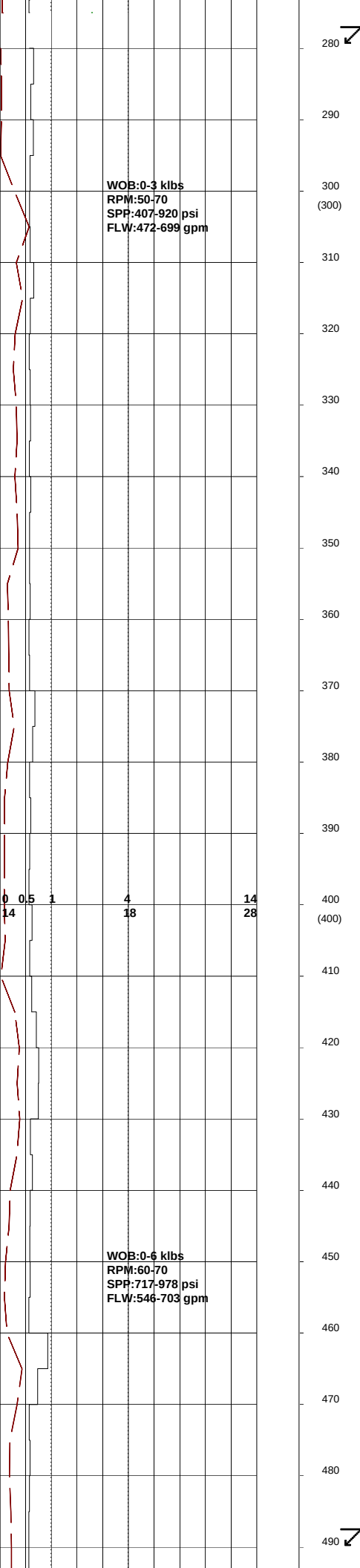
NO GAS

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

21-12-2017

22-12-2017

14 1 4 14
18 28



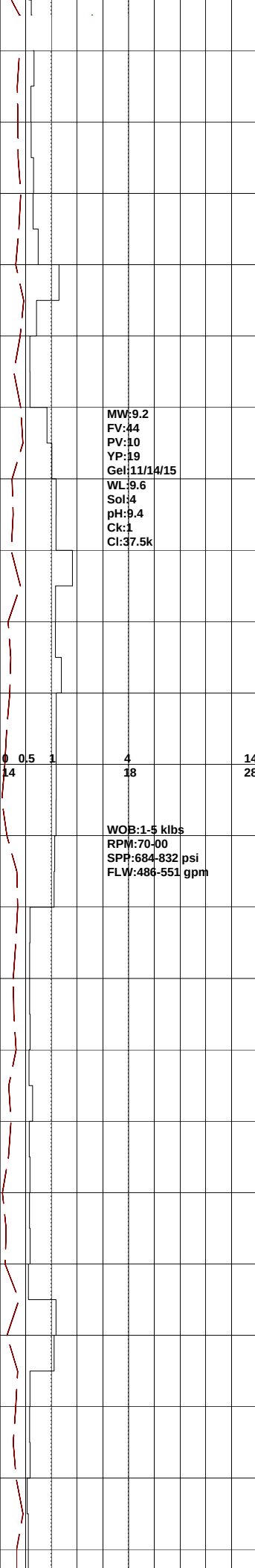
SURVEY @ 276.94°MDRT
INC: 0.09° AZI: 102.14°G

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

NO GAS

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

SURVEY @ 487.39°MDRT
INC: 0.18° AZI: 345.94°G

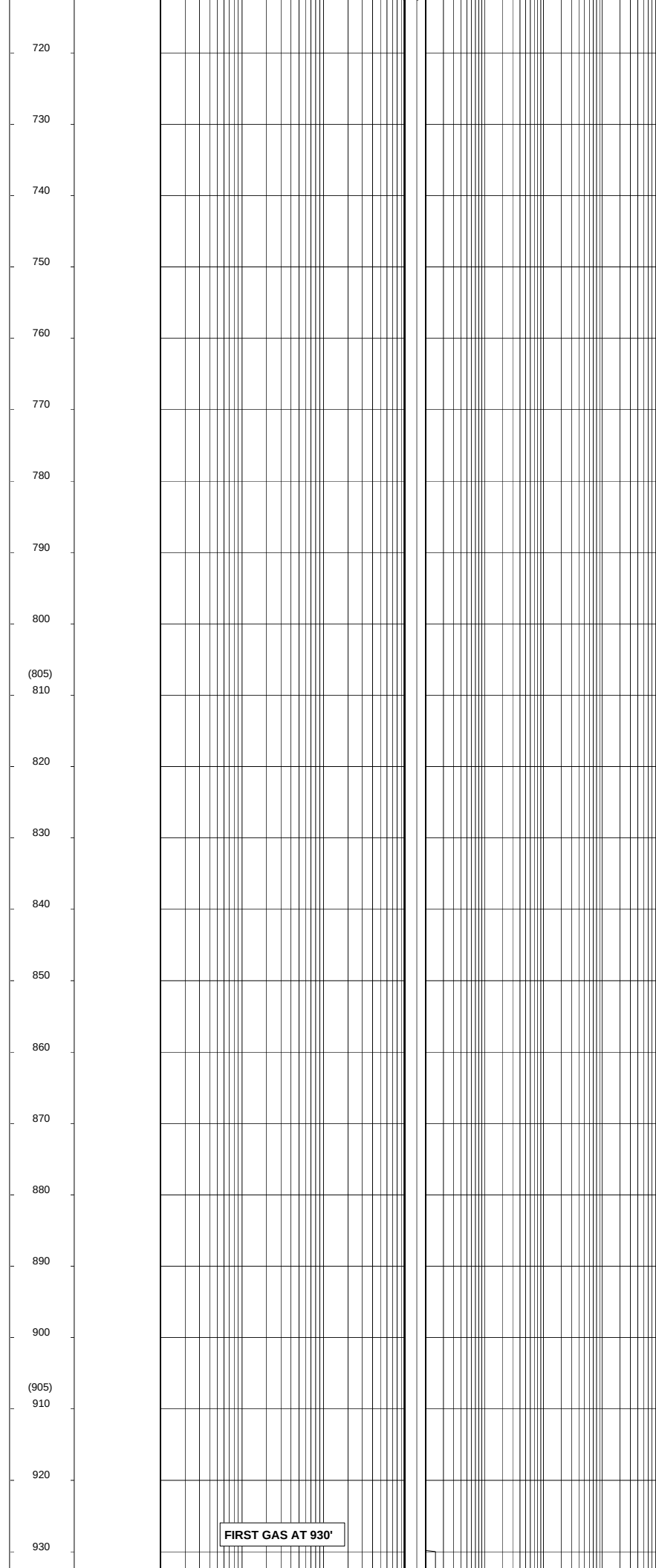
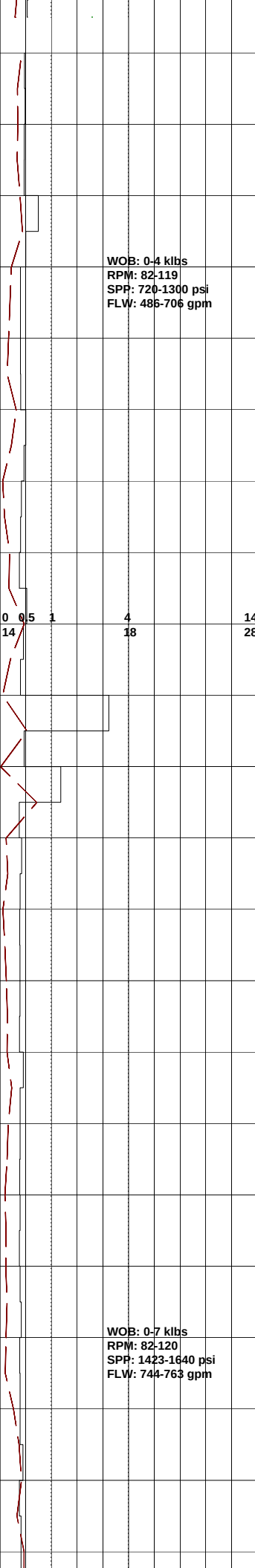


NO GAS			
500			
(505)			
510			
520			
530			
540			
550			
560			
570			
580			
590			
600			
(605)			
610			
620			
630			
640			
650			
660			
670			
680			
690			
700			
(705)			
710			

SURVEY @ 581.61'MDRT
INC: 0.18° AZI: 297.22°G

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

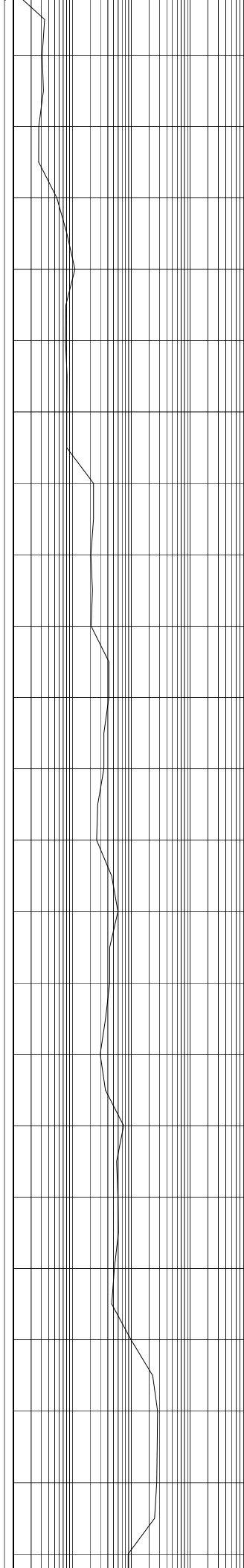
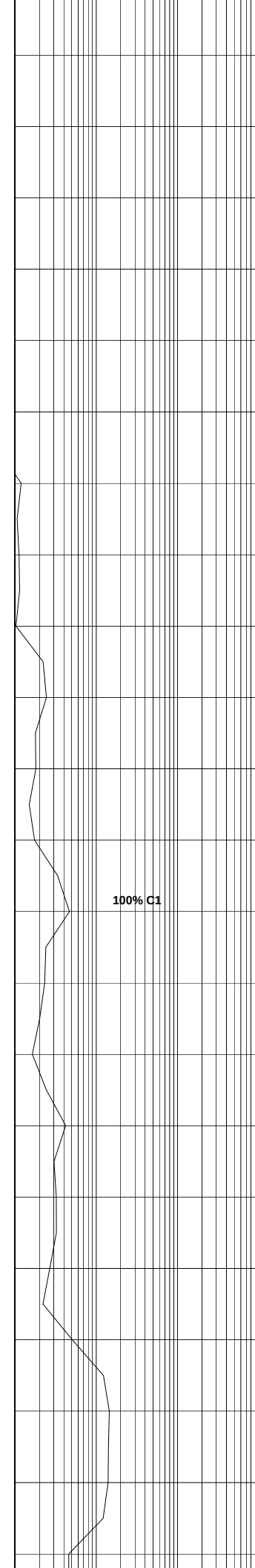
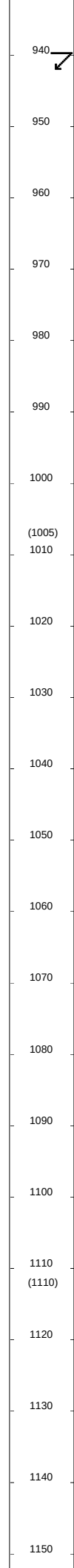
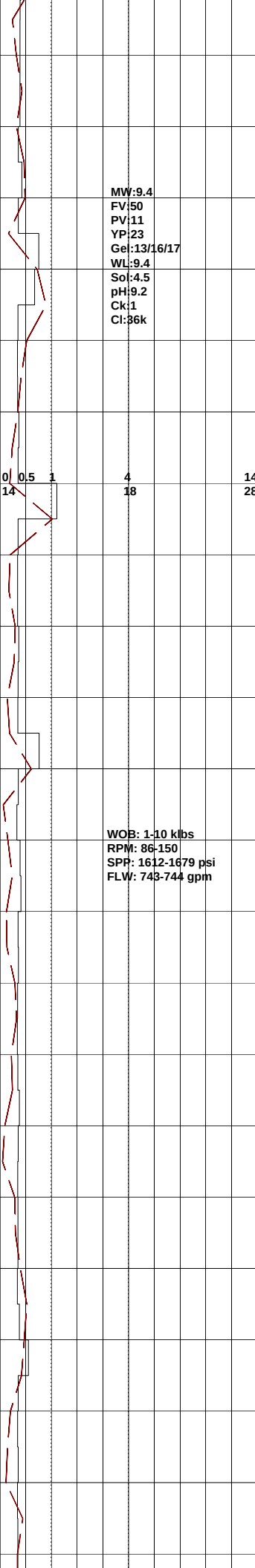
SURVEY @ 671.30'MDRT
INC: 0.46° AZI: 355.64°G



NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

FIRST GAS AT 930'

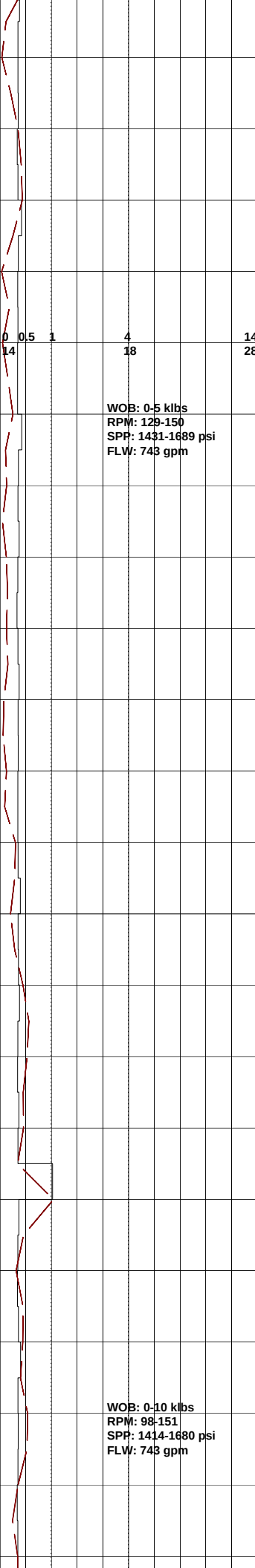


SURVEY @ 939.38' MDRT
INC: 0.33° AZI: -

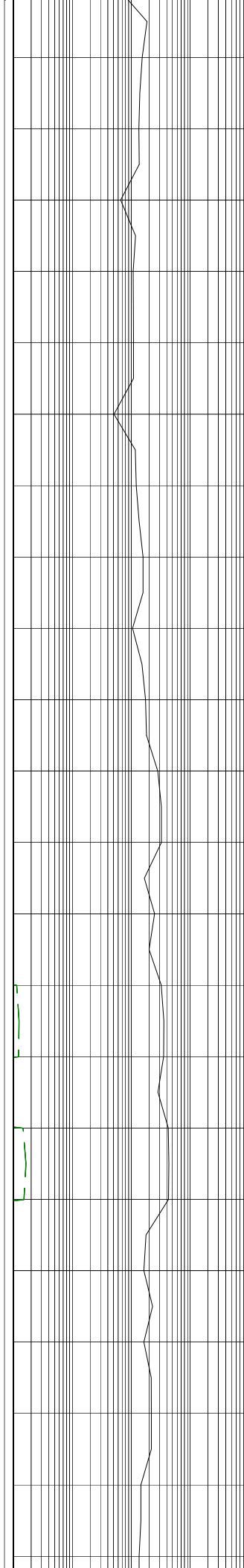
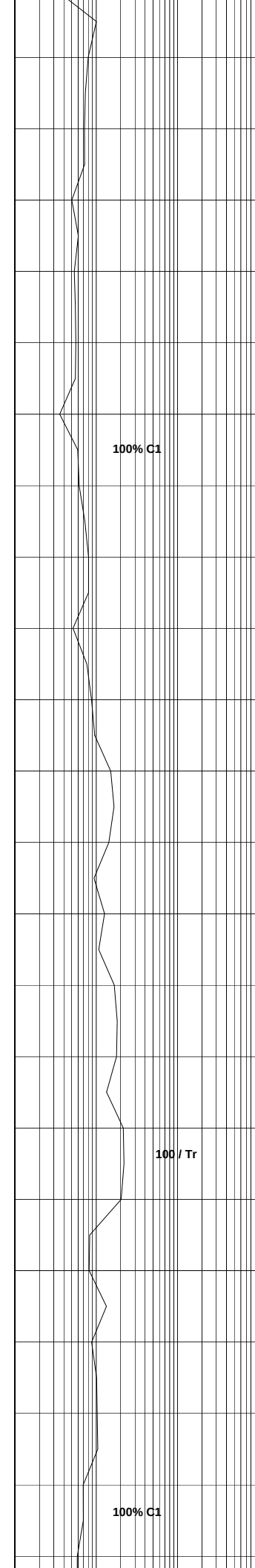
AZIMUTH NOT AVAILABLE
FOR SURVEYS 939'-2254'

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

100% C1



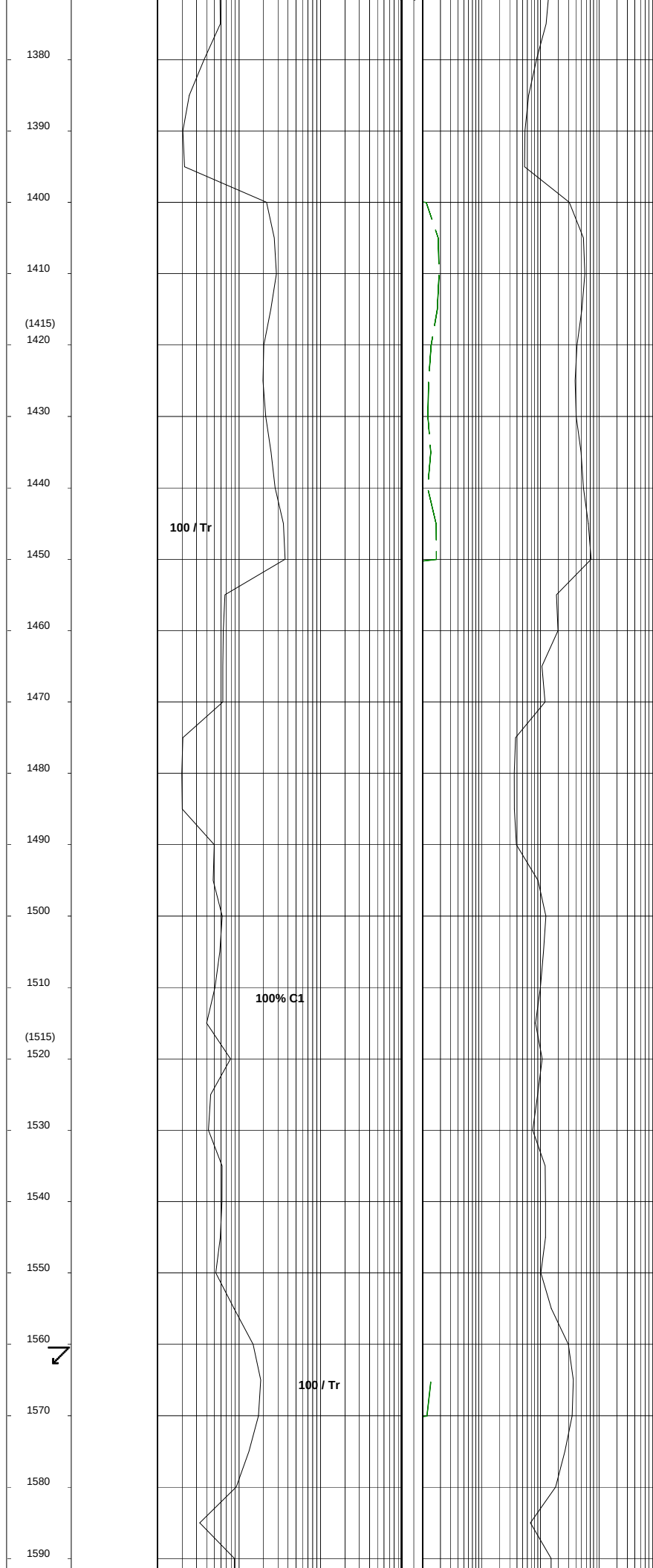
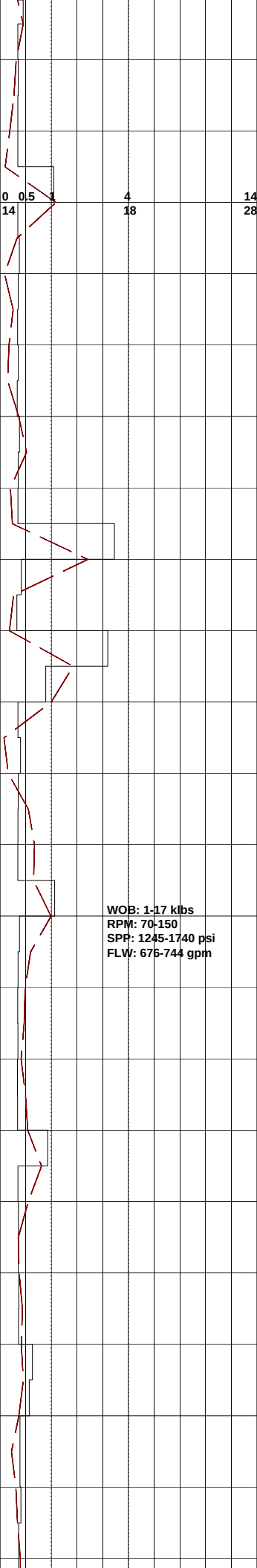
1160
1170
1180
1190
1200
1210
(1210)
1220
1230
1240
1250
1260
1270
1280
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1300
1310
(1310)
1320
1330
1340
1350
1360
1370

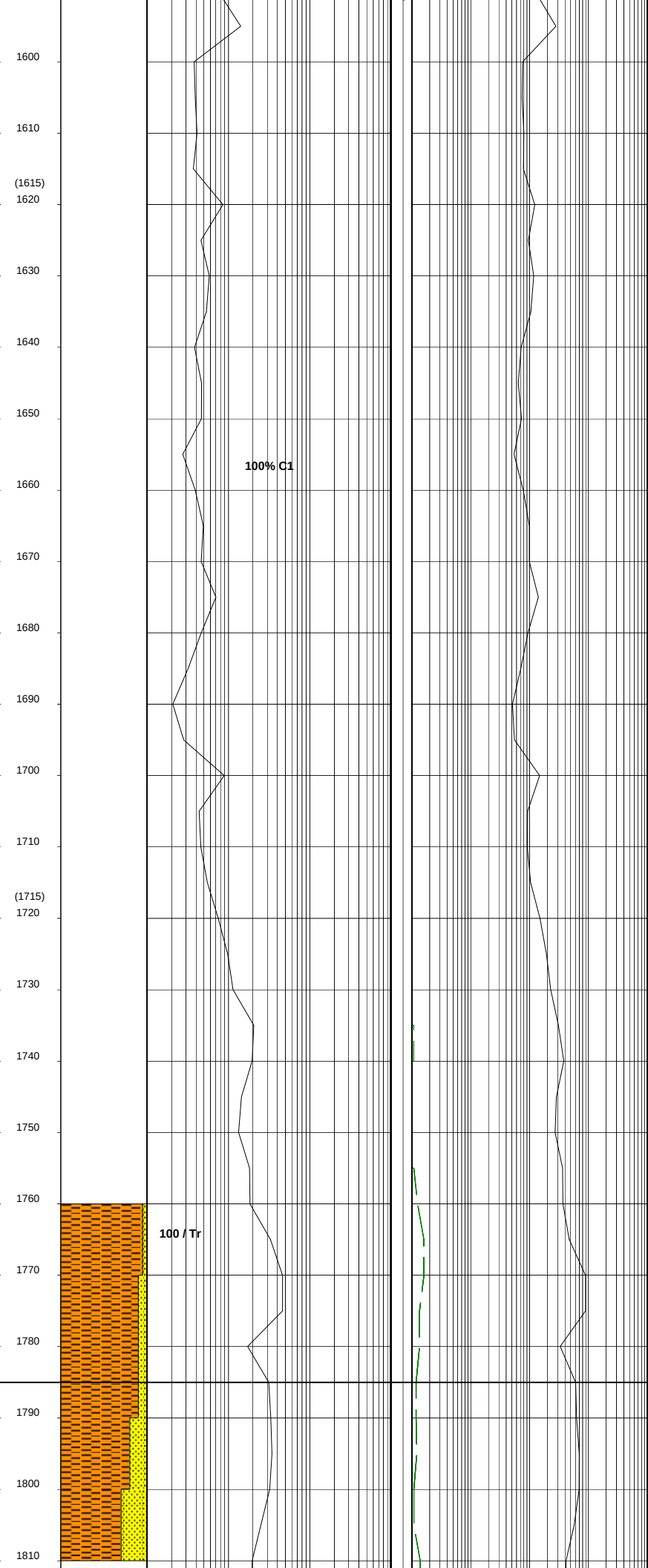
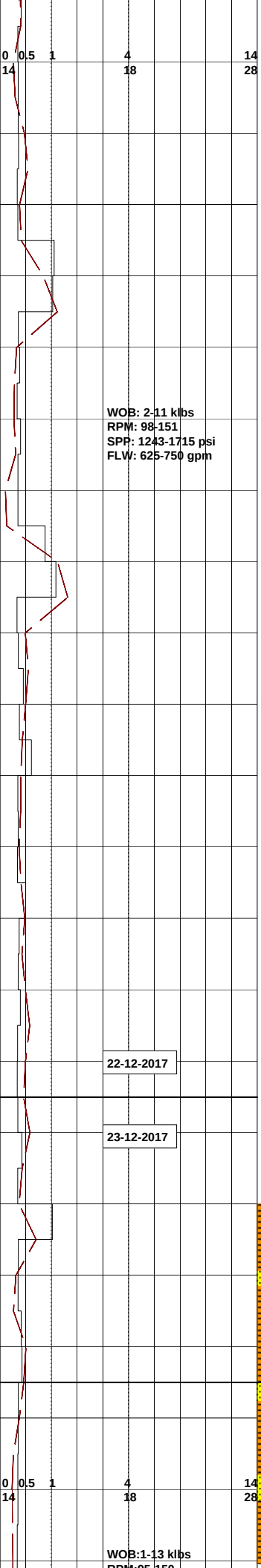


NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

SURVEY @ 1249.86'MDRT
INC: 0.44° AZI: -

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE





NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

MACKUNDA FORMATION:
1785' MDRT (-1635' TVDSS)

RPM:95-130
SPP:1712-1794 psi
FLW:748-750 gpm

WOB:1-13 klbs
RPM:82-149
SPP:1385-1807 psi
FLW:646-748 gpm

0 0.5 1 4 14
14 18 28

(1815)
1820

1830

1840

1850

1860

1870

1880

1890

1900

1910

1920
(1920)

1930

1940

1950

1960

1970

1980

1990

2000

2010

2020
(2020)

2030

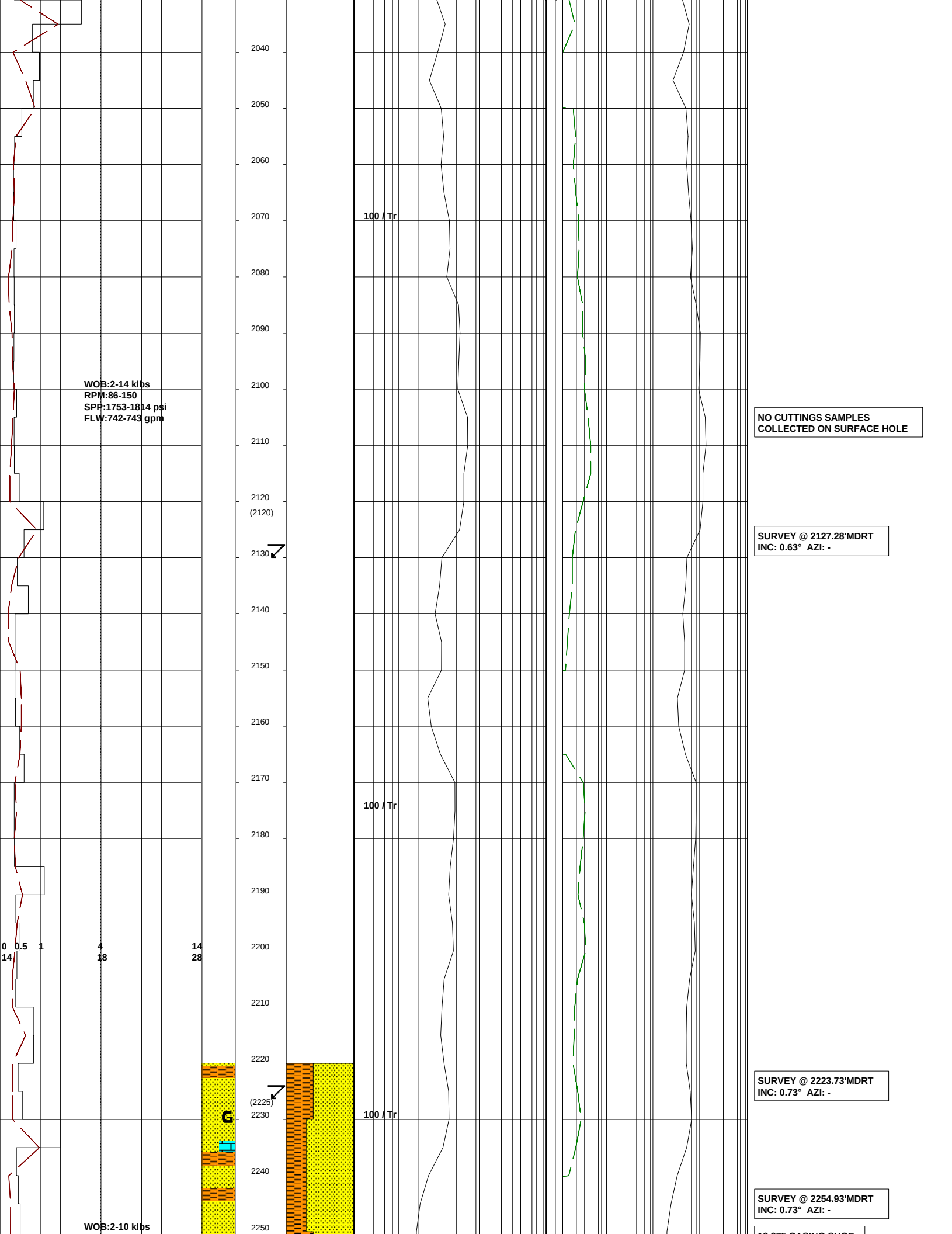
100 / Tr

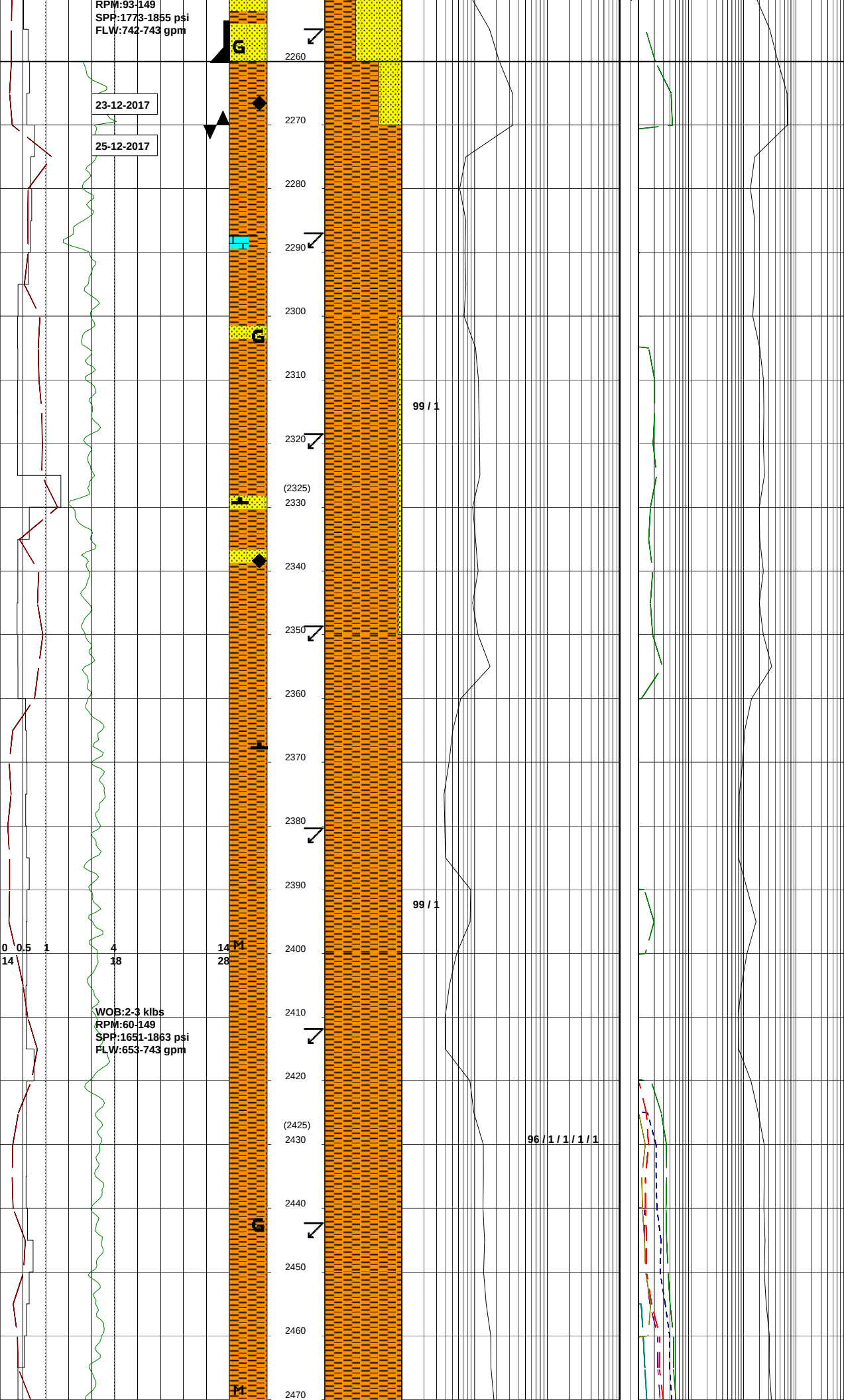
100 / Tr

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE

SURVEY @ 1878.0' MDRT
INC: 0.53° AZI: -

NO CUTTINGS SAMPLES
COLLECTED ON SURFACE HOLE





13.3/5 CASING SHOE
SET @ 2260'MDRT

OODNADATTA FORMATION:
2260'MDRT (-2110'TVDSS)

BIT #2 SMITH S519HEPX
SIZE:12.25" JETS:6x13,1x15
IN:2270' OUT:5077'
RUN:2807' HRS:28.5
COND:1-1-DL-IN-X-I-RR-TD

SURVEY @ 2286.39'MDRT
INC: 1.18° AZI: 74.55°G

LIMESTONE:(Tr) pl orng,off wh,
tan,lut,micrxln,brit,mod hd.

SILTSTONE:med gry,arg,rr
carb spks,tr micmic,tr calc frags,
rr dissem pyr,frm-occ mod hd,sbblky.

SURVEY @ 2318.61'MDRT
INC: 1.87° AZI: 63.04°G

SANDSTONE:lt gry,off wh,
f-vf,wl srt,sbrnd-sbang,wk calc
cmt,com lt gry slty mtx,rr off wh
arg mtx,occ carb spks,rr glauc grs,
tr pyr,fri aggs,pr vis por,no fluor.

SURVEY @ 2349.11'MDRT
INC: 2.85° AZI: 56.38°G

SILTSTONE:med-lt brnsh gry,
med gry,arg,rr aren,mnr carb spks
& frags,tr liths,rr calc frags,frm-
occ mod hd,sbblky-occ blkly.

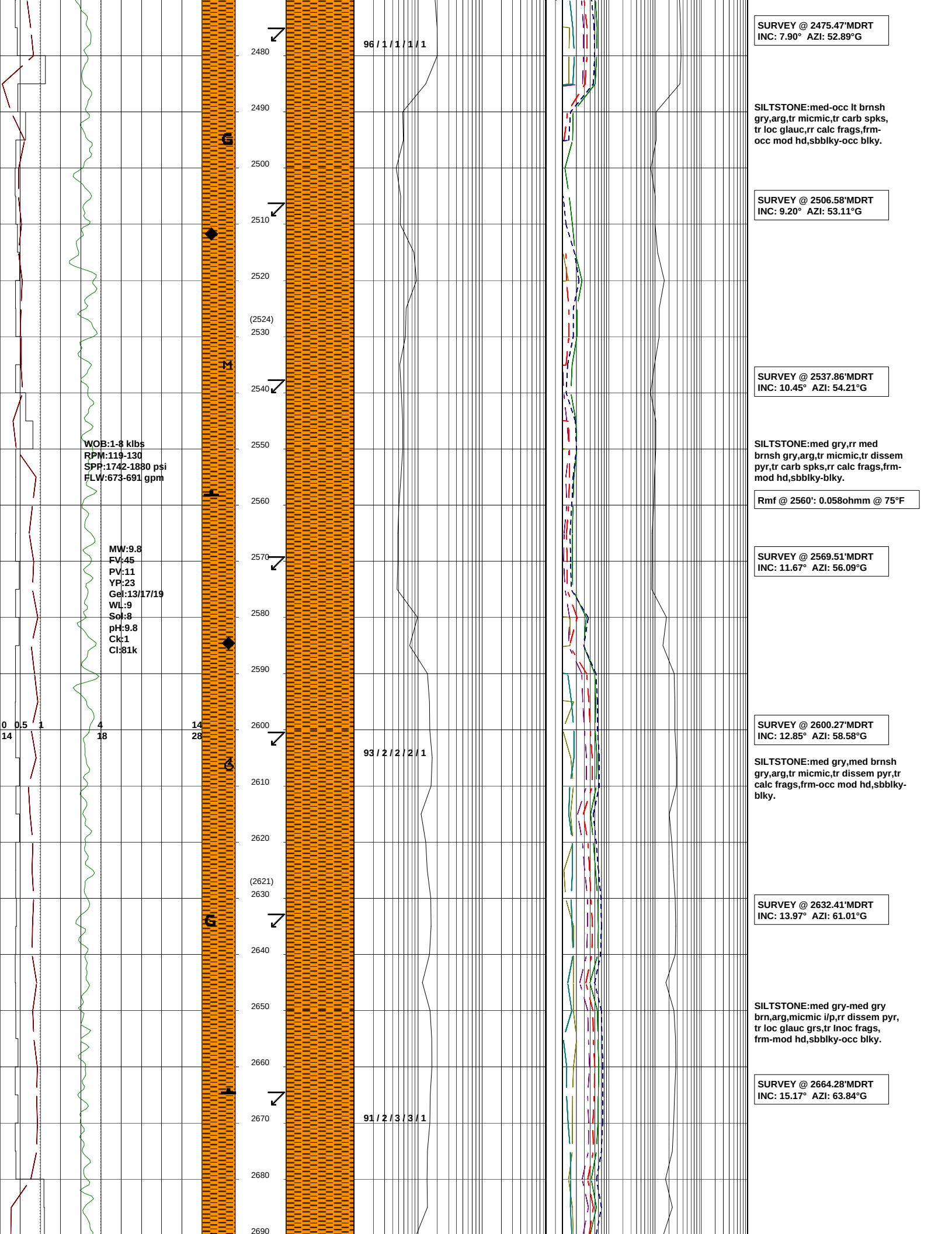
SURVEY @ 2380.40'MDRT
INC: 3.96° AZI: 54.36°G

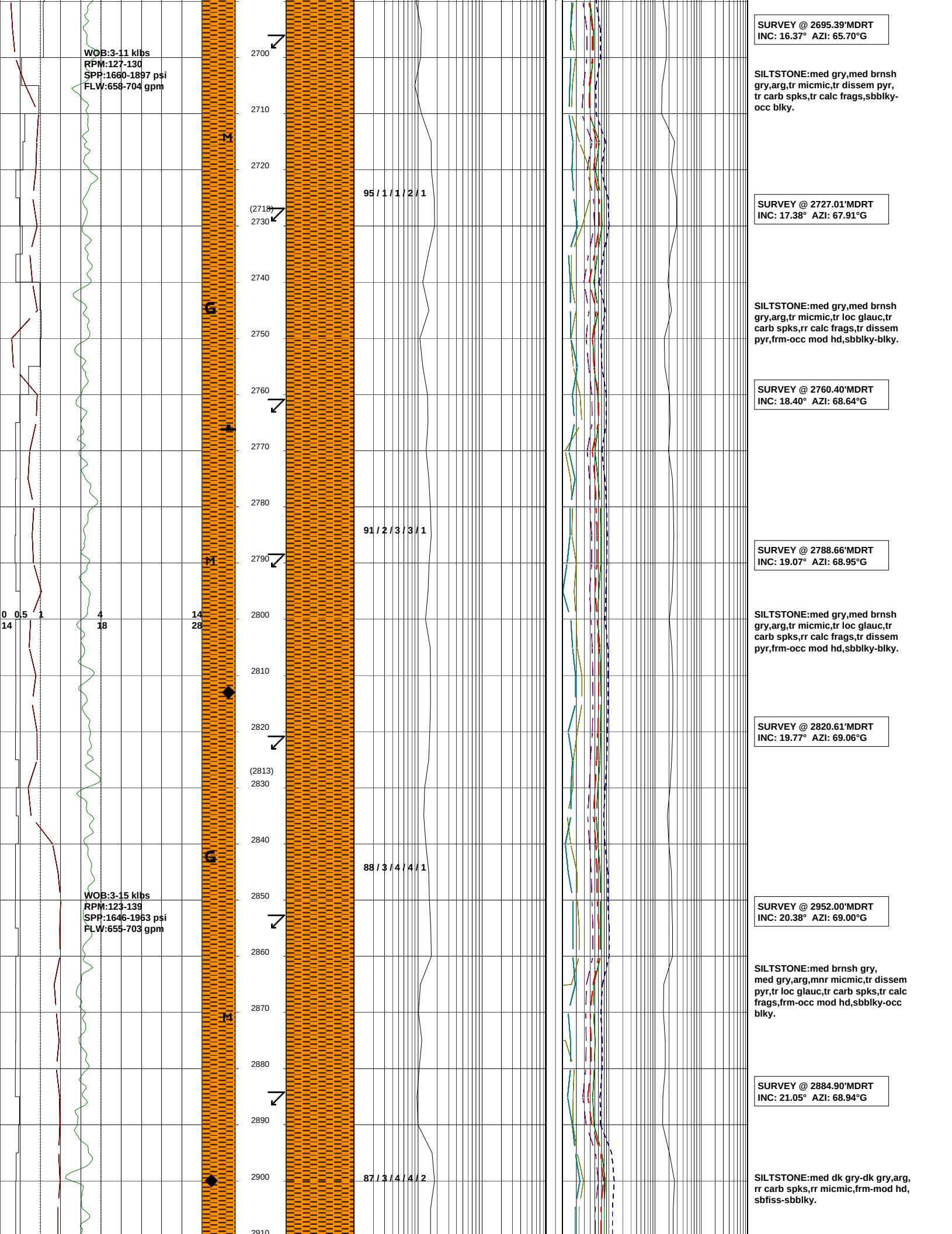
SILTSTONE:med-rr lt gry,
lt brnsh gry,arg,tr carb spks,
tr micmic,rr calc frags,frm-
occ mod hd,sbblky-occ blkly.

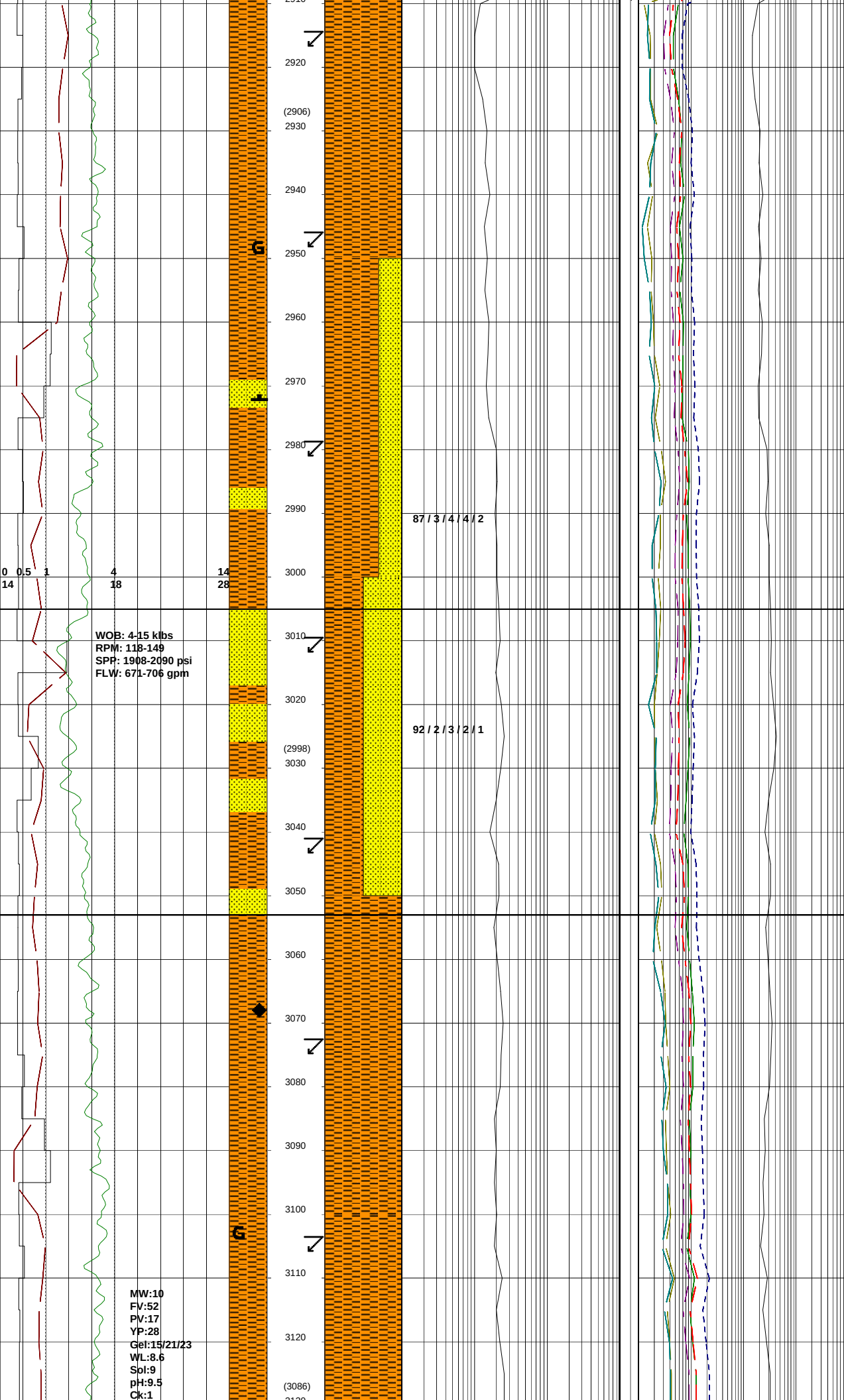
SURVEY @ 2411.49'MDRT
INC: 5.21° AZI: 54.07°G

SURVEY @ 2442.57'MDRT
INC: 6.50° AZI: 53.53°G

SILTSTONE:med gry,med brnsh
gry,arg,rr loc glauc,tr micmic,tr carb
spks,frm-occ mod hd,sbblky-blky.







SURVEY @ 2914.59'MDRT
INC: 21.65° AZI: 68.83°G

SILTSTONE:med gry-occ med gry
brn,dm arg-occ aren,micmic,loc
com glauc grs,mnr felds,mod hd,v
pr-ti vis por,no fluor.

SURVEY @ 2946.33'MDRT
INC: 22.41° AZI: 68.75°G

SANDSTONE:off wh-lt gry,mnr med
grn,vf-f,wl srt,sbang-sbrnd,mod
calc cmt,com-abdt off wh-lt gry
arg mtx,abdt glauc grs,mnr felds,
mod hd,v pr-ti vis por,no fluor.

SURVEY @ 2978.24'MDRT
INC: 23.61° AZI: 67.73°G

SURVEY @ 3009.86'MDRT
INC: 24.9° AZI: 67.73°G

COORIKIANA SANDSTONE:
3005'MDRT (-2830'TVDSS)

SILTSTONE:med-lt gry,arg-rr aren,
micmic,tr carb spks,mod hd-frn,
sbbiky-occ blkly.

SANDSTONE:off wh-lt gry,f,occ vf-
rr med,pr srt,sbang-mnr sbrnd,mod-
strg calc cmt,com lt gry arg mtx,
com-abdt glauc grs,mnr liths & carb
spks,mod hd-occ hd,no fluor.

SURVEY @ 3040.85'MDRT
INC: 26.17° AZI: 67.24°G

BULLDOG SHALE:
3053'MDRT (-2873'TVDSS)

SILTSTONE:med gry brn,med gry,
arg,tr aren,rr carb spks,rr dissem
pyr,frm-tr mod hd,sbbiky-blkly.

SURVEY @ 3072.07'MDRT
INC: 27.28° AZI: 66.59°G

SURVEY @ 3103.57'MDRT
INC: 28.49° AZI: 66.22°G

SILTSTONE:med gry brn,med gry,
arg,tr aren,rr carb spks,rr dissem
pyr,rr glauc,frm-tr mod hd,sbbiky-
blkly.

WOB: 4-15 klbs
RPM: 118-149
SPP: 1908-2090 psi
FLW: 671-706 gpm

MW:10
FV:52
PV:17
YP:28
Gel:15/21/23
WL:8.6
Sol:9
pH:9.5
Ck:1

81 / 4 / 6 / 7 / 2

SURVEY @ 3135.33'MDRT
INC: 29.68° AZI: 65.69°G

SILTSTONE:med gry,occ med gry
brn,dom arg-occ aren,micmic,tr
carb spks,mnr glauc grs,mod hd-
occ hd,sbbiky-blky.

SURVEY @ 3167.43'MDRT
INC: 30.68° AZI: 65.31°G

SILTSTONE: med gry-occ med gry
brn,arg,micmic,occ calc incl,mod
hd-occ hd,blky-sbblky.

SURVEY @ 3199.01'MDRT
INC: 31.76° AZI: 65.24°G

SURVEY @ 3229.52'MDRT
INC: 32.57° AZI: 65.01°G

SILTSTONE: med gry, med brnsh gry, tr med dk blush gry, arg, rr micmic, rr carb spks, mod hd, sbblky-blky.

SURVEY @ 3261.73'MDRT
INC: 33.54° AZI: 64.95°G

SURVEY @ 3291.90'MDRT
INC: 34.43° AZI: 65.01°G

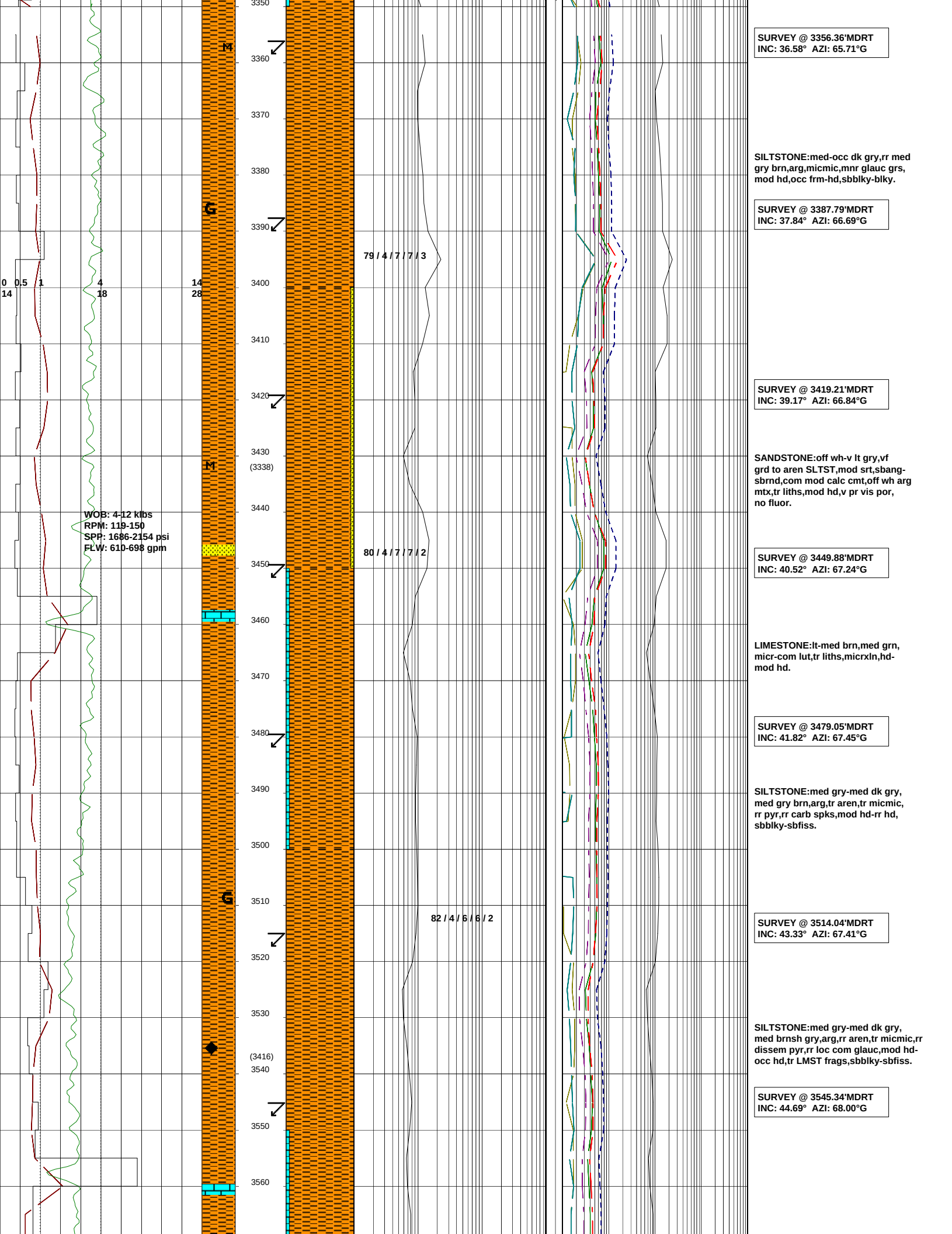
SILTSTONE: med gry, med brnsh gry, tr med dk gry, arg, rr micmic, rr carb spks, mod hd, sbblky-blky.

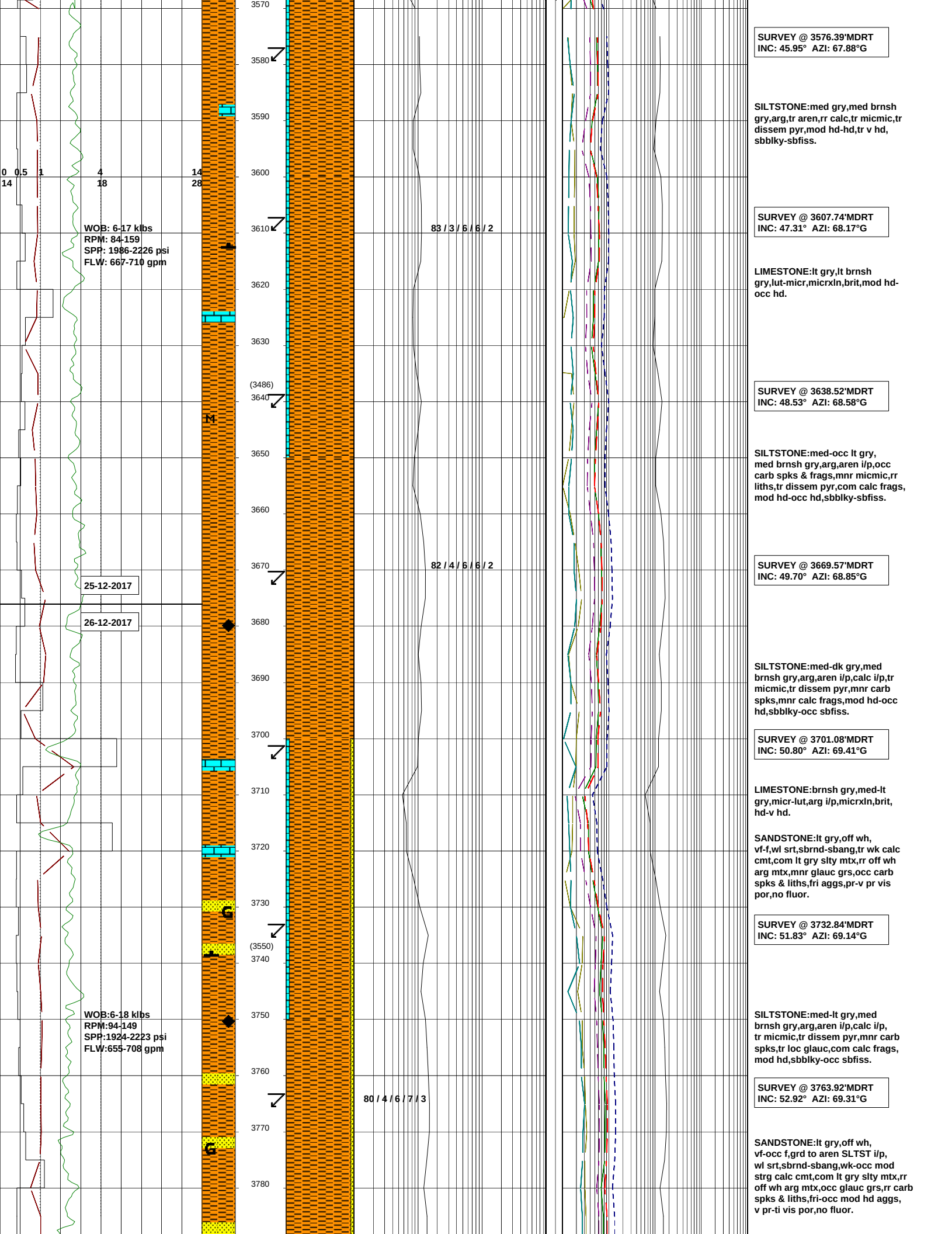
SURVEY @ 3324.98'MDRT
INC: 35.42° AZI: 65.22°G

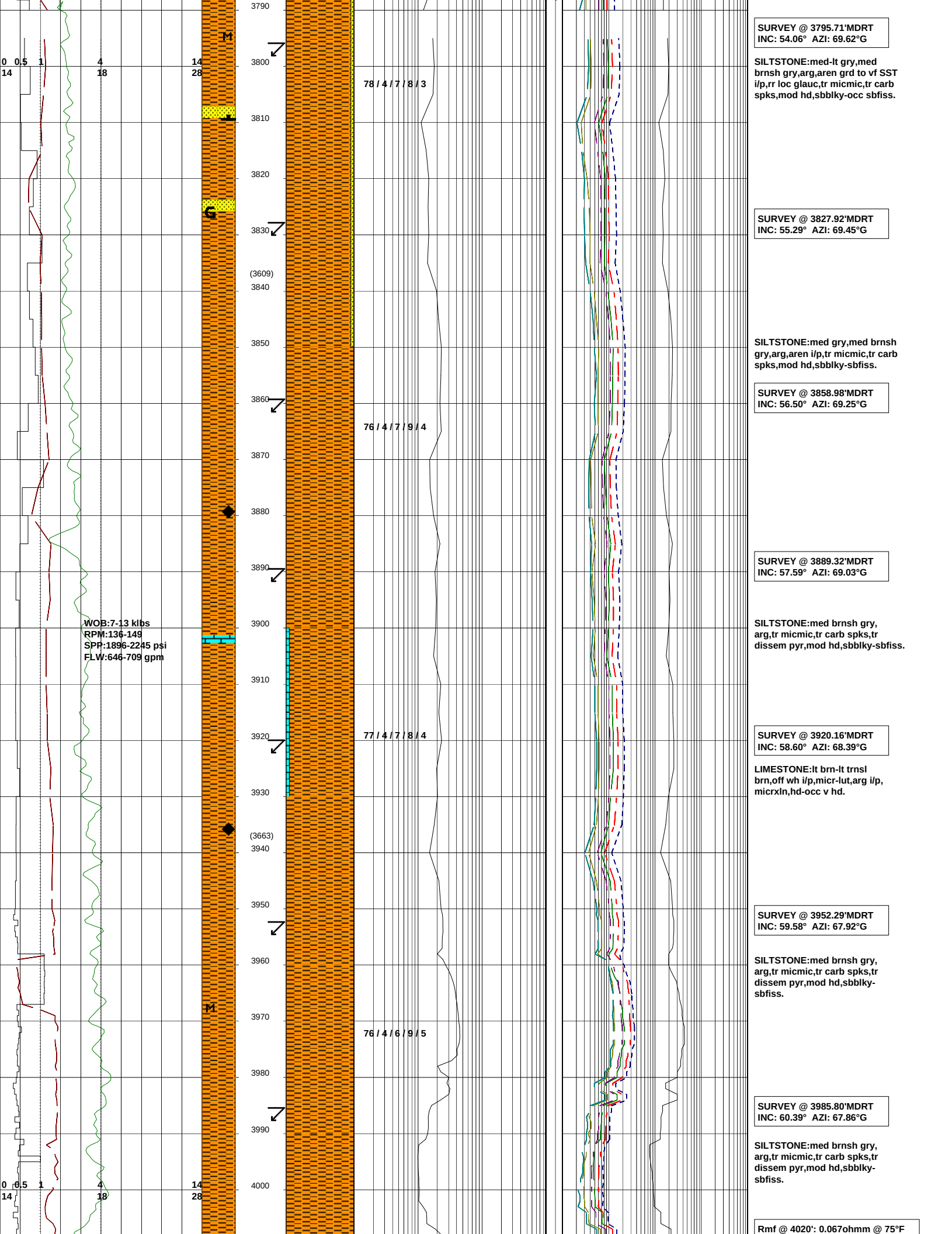
LIMESTONE:off wh-lt trnsl gry,
lt trnsl brn,micr-com lut,tr liths,
micrxln-xln,hd-mod hd.

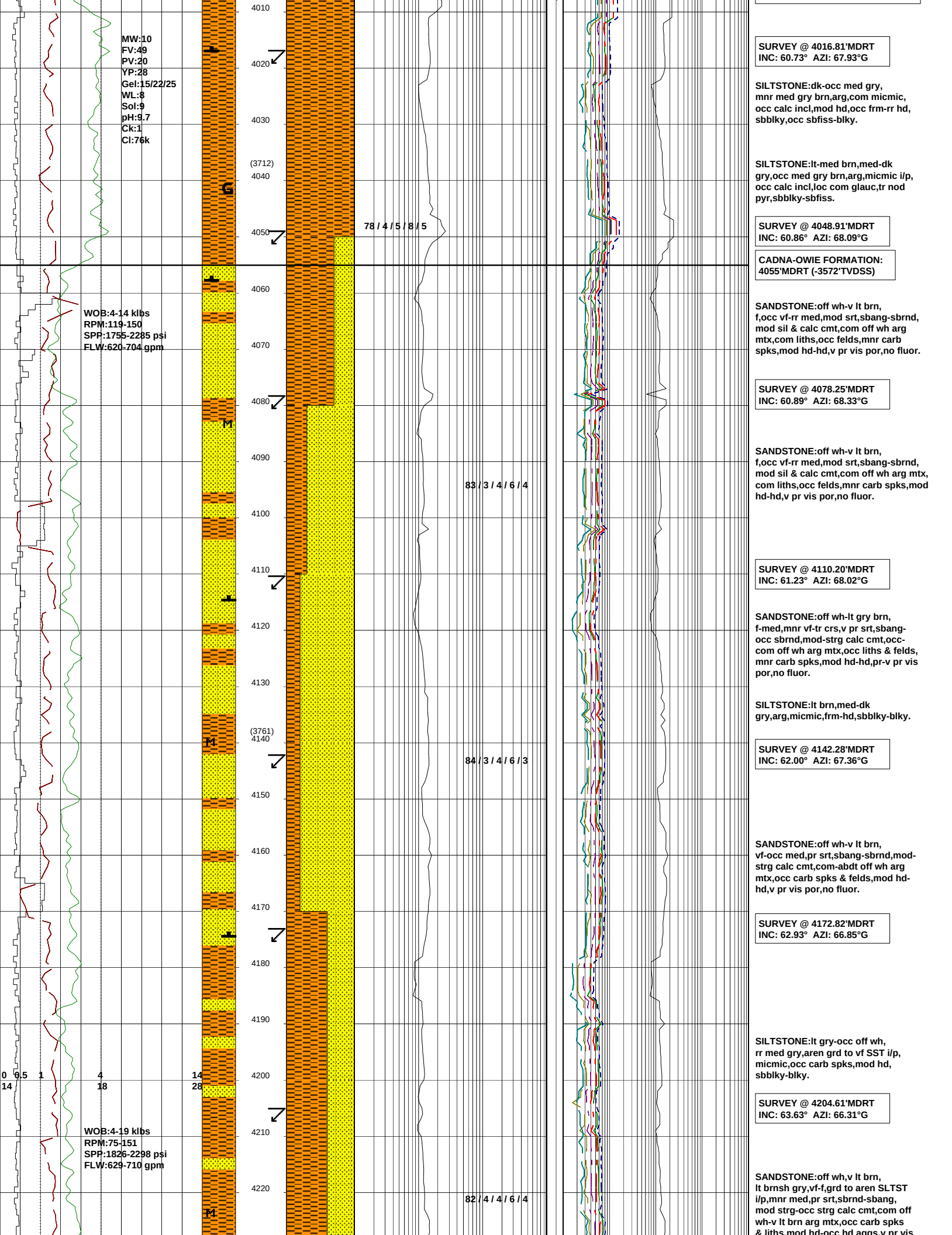
WOB: 3-14 klbs
RPM: 88-150
SPP: 1822-2114 psi
FLW: 648-705 gpm

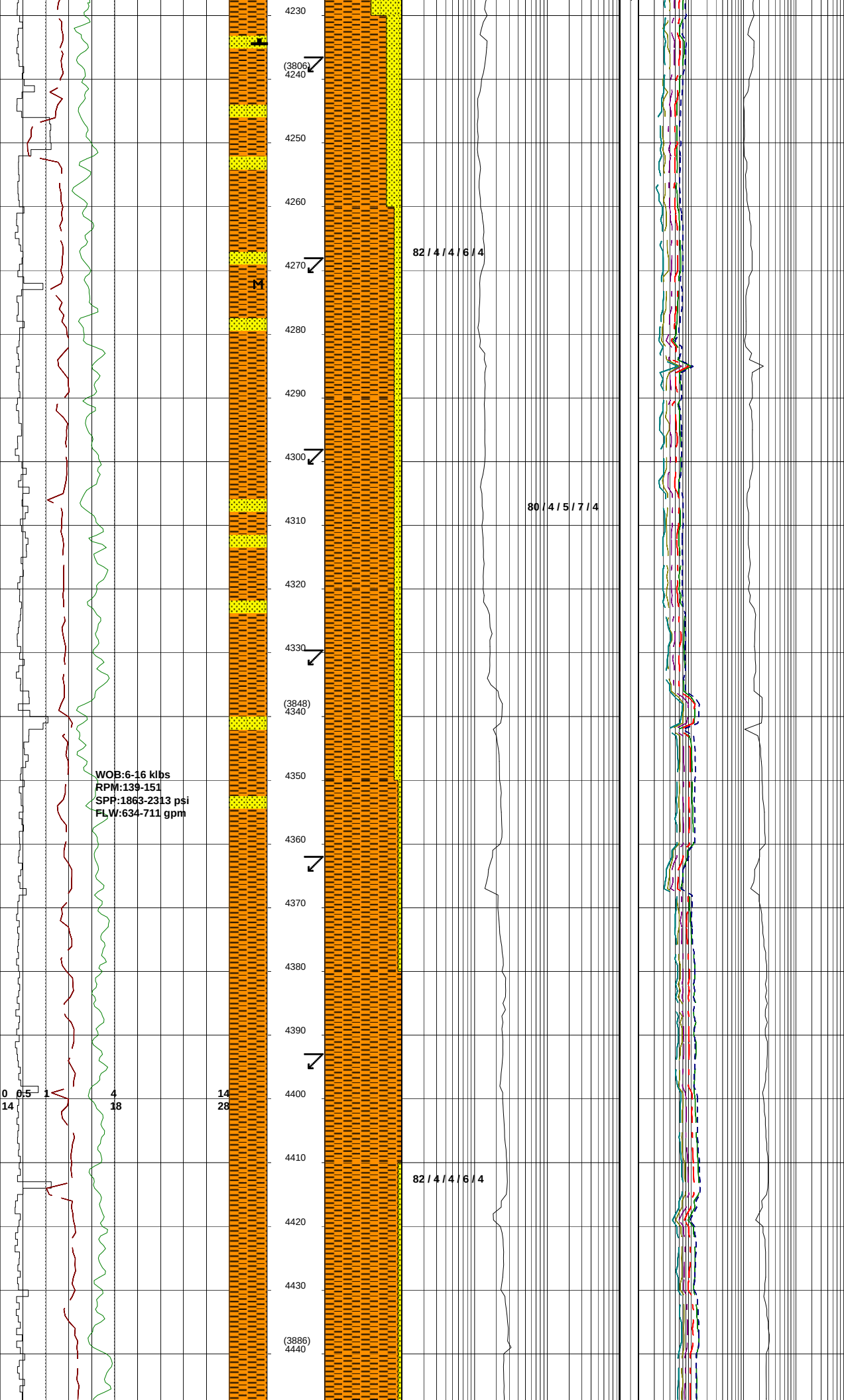
WOB: 6-13 klbs
RPM: 147-150
SPP: 1680-2158 psi
FLW: 612-704 gpm

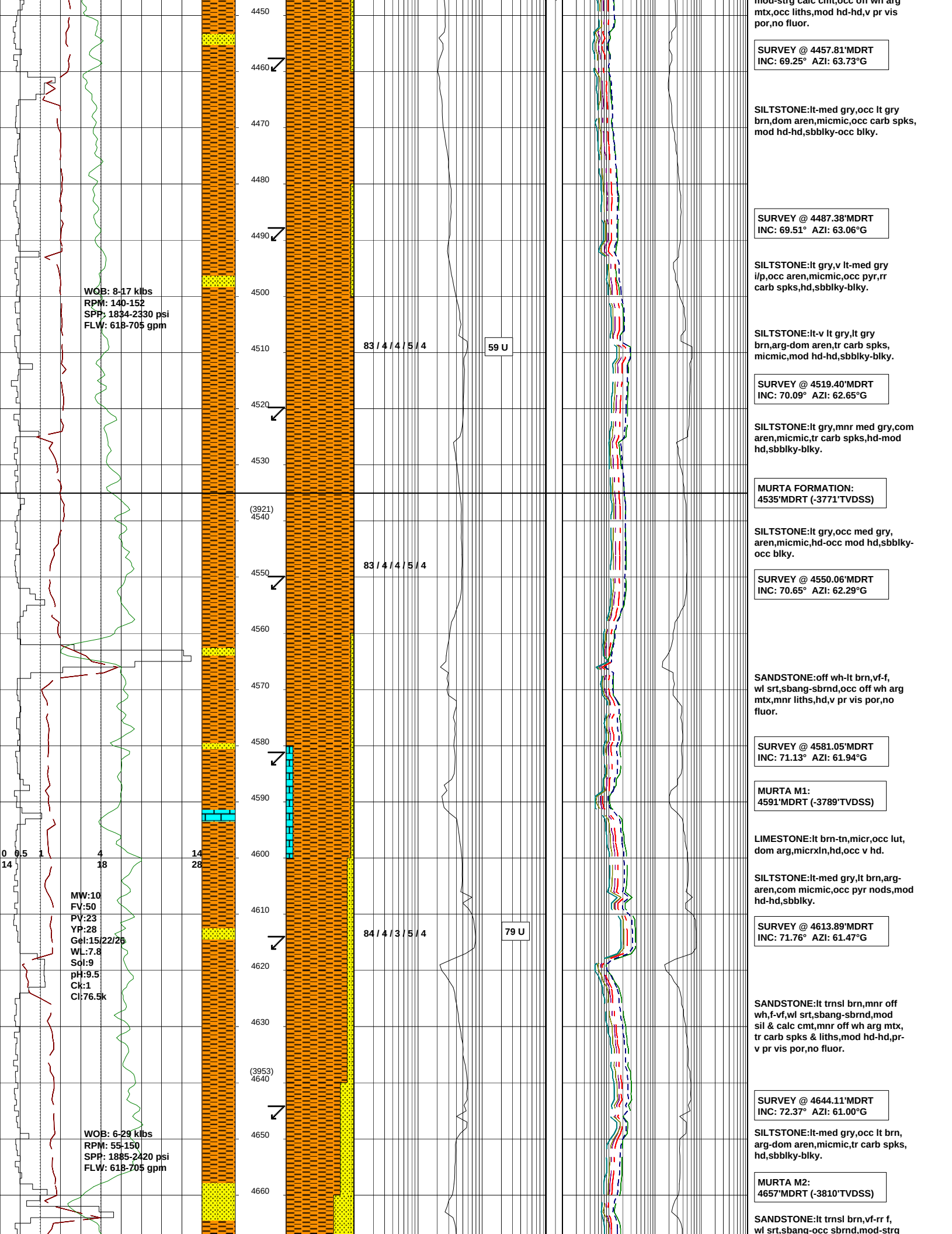


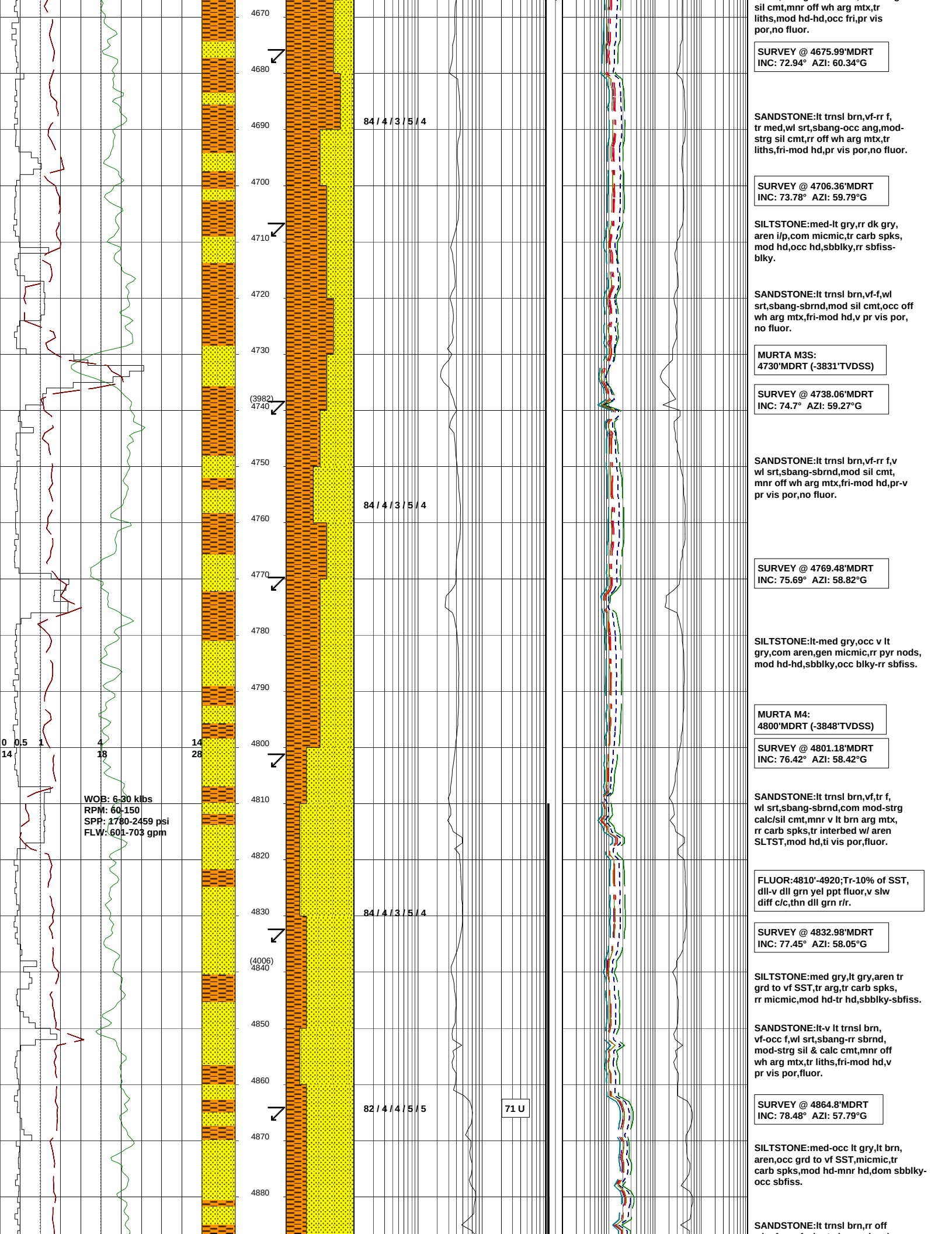


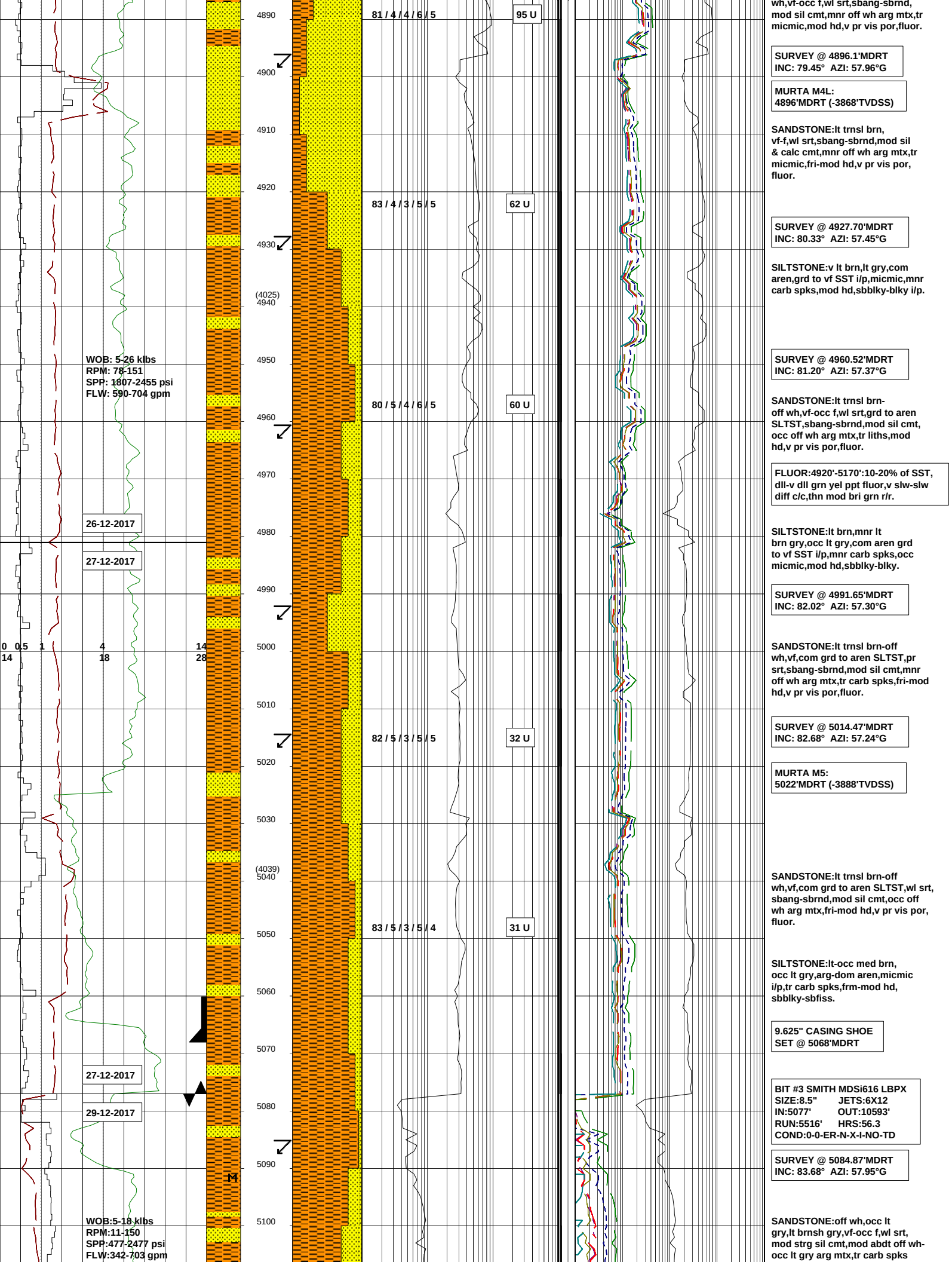








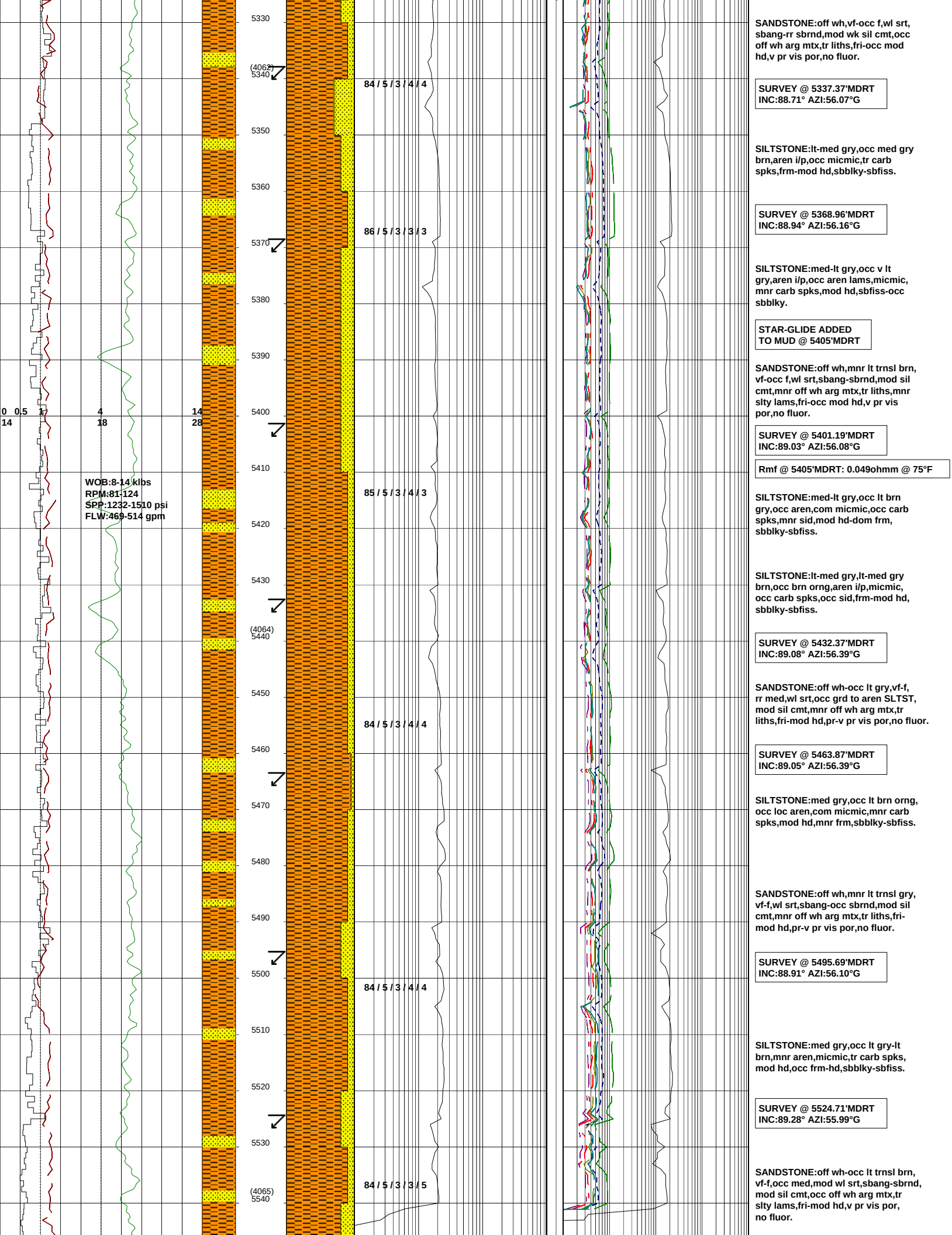


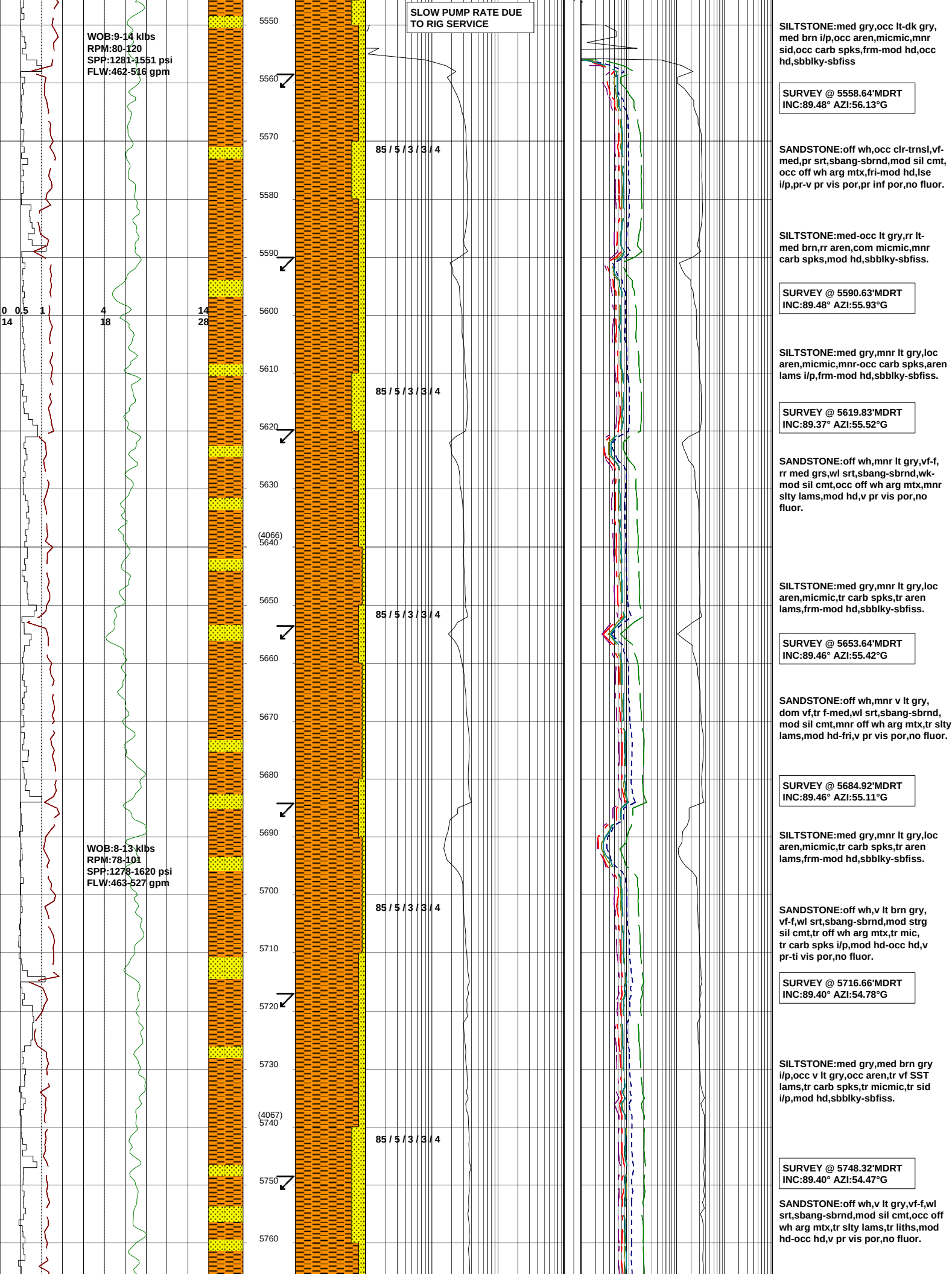


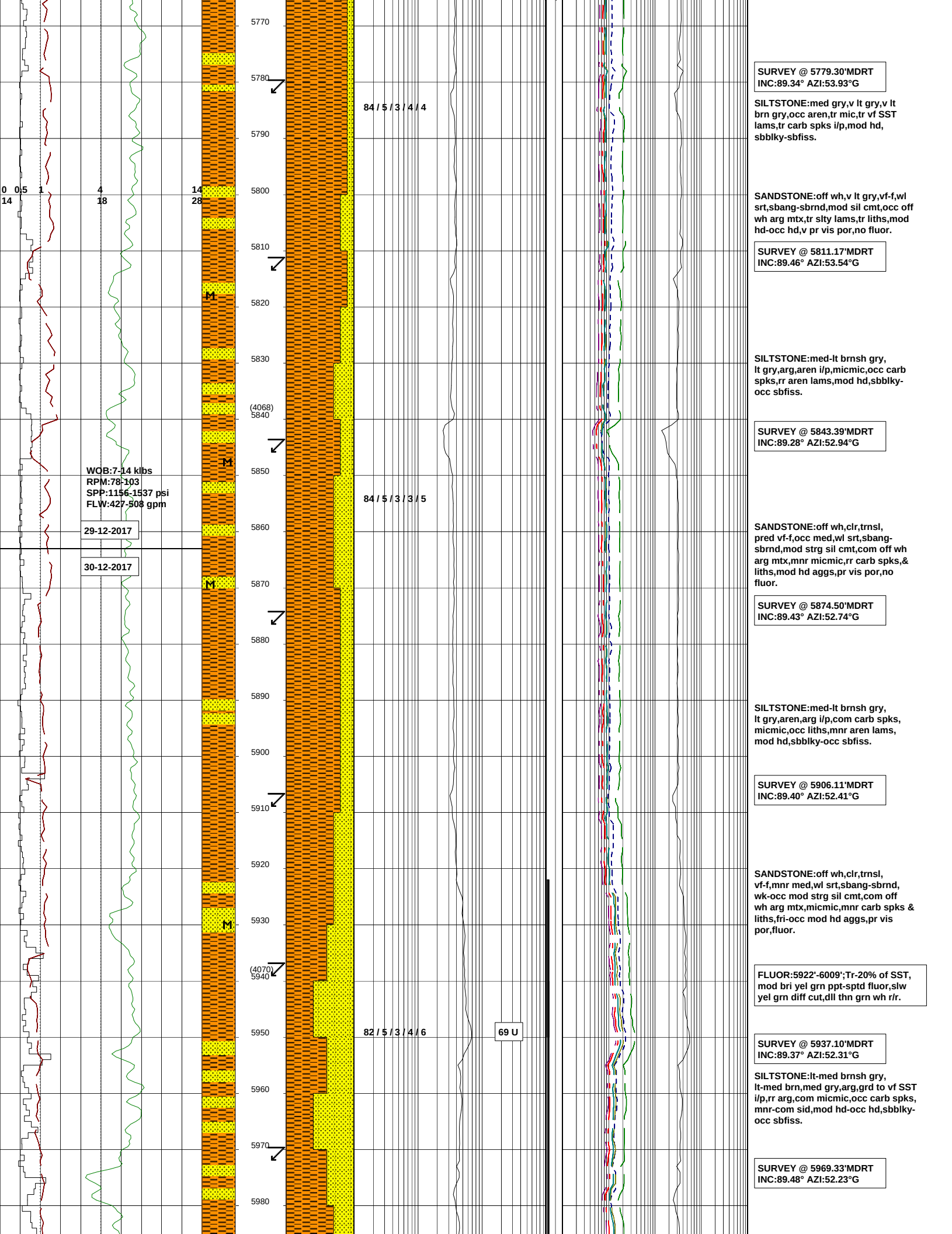
SILTSTONE;med-occ lt gry,aren i/p,
com micmic,rr-tr carb spks,frm-occ
mod hd,sbblky-sbfi ss.

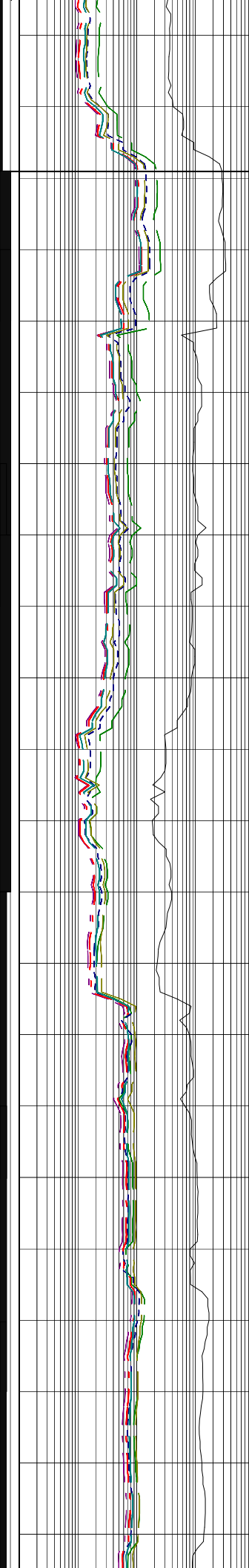
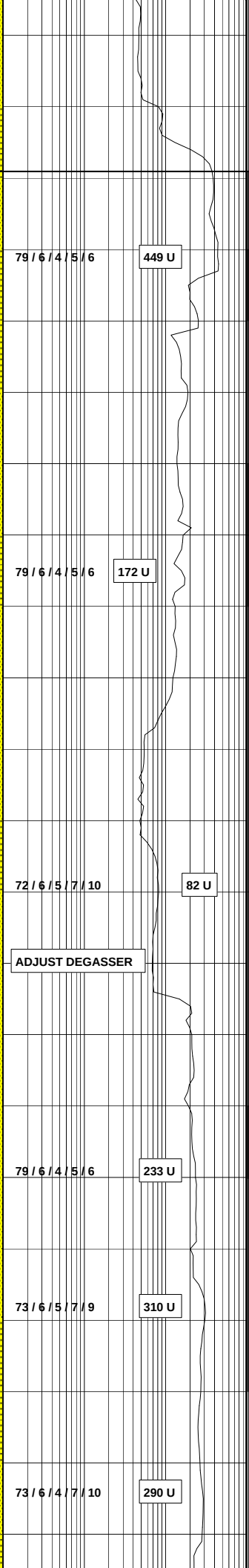
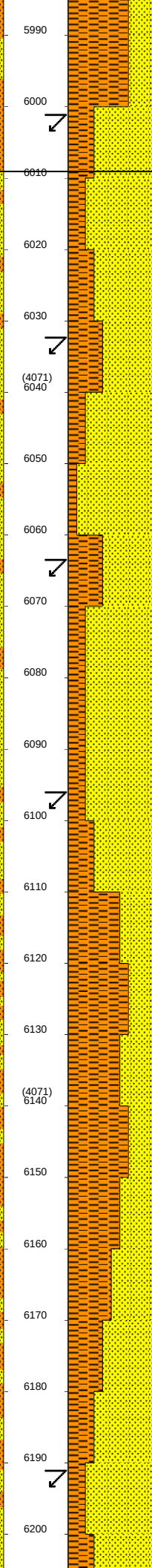
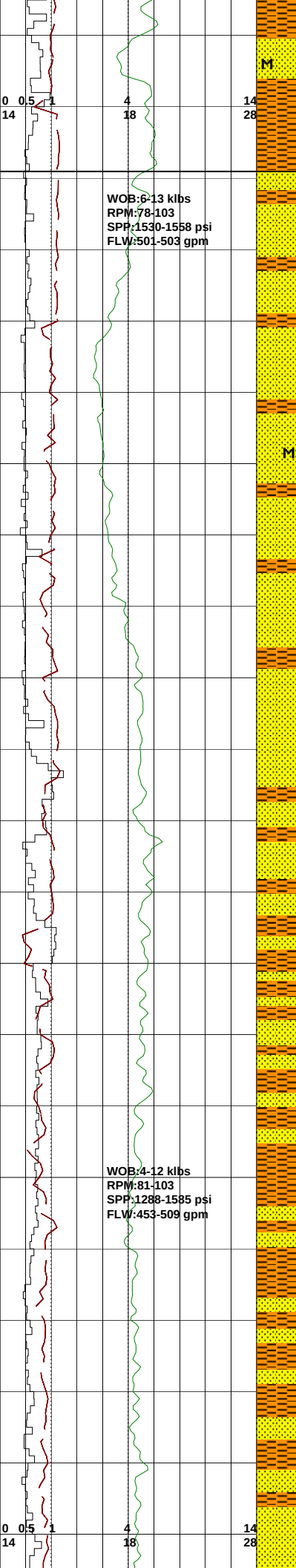
WOB:4-15 klbs
RPM:80-124
SPP:1160-1597 psi
FLW:462-541 gpm

84 / 5 / 3 / 4 / 4









SURVEY @ 6001.03°MDRT
INC: 89.23° AZI: 51.77°G

MCKINLAY MEMBER:
6009°MDRT (-3920°TVDSS)

SANDSTONE: off wh, clr, trnsi, opq, pred med-crs, mnrf, pr-mod wl srt, sbang-sbrnd, mod strg sil cmt, mnrf off wh arg mtz, occ mic flks, pred cln lse grs, mnrf grn aggs, fr-gd inf por, pr vis por, fluor.

FLUOR: 6009°-6070°; 40-70% of SST, dll-mod bri yel grn ptchy-evn fluor, slw yel grn diff cut, thk grn wh rlr.

SURVEY @ 6032.74°MDRT
INC: 89.26° AZI: 51.92°G

SANDSTONE: clr, trnsi, opq, off wh, med-crs, pr srt, sbang-sbrnd, tr wk-mod strg sil cmt, mnrf off wh arg mtz, occ mic flks, pred cln lse qtz grs, fri-mod hd aggs, gd-fr vis por, pr-fr inf por, fluor.

SURVEY @ 6063.98°MDRT
INC: 89.43° AZI: 51.87°G

FLUOR: 6070°-6110°; 60-90% of SST, mod bri yel grn evn fluor, slw-mod rapid strmg cut, thk flm res.

SANDSTONE: clr-trnsi, lt trnsi brn, rr off wh, f-med, occ crs-v crs, pr srt, ang-occ sbrnd, wk-mod strg sil cmt, occ off wh arg mtz, mnrf pyr nodes, rr qtz ovghs, mnrf slty lams, lse, fri-mod hd, fr inf por, pr-fr vis por, fluor.

SURVEY @ 6096.06°MDRT
INC: 89.66° AZI: 52.06°G

SANDSTONE: clr-trnsi, lt trnsi brn, rr off wh, f-med, rr crs, pr srt, ang-occ sbrnd, wk-mod strg sil cmt, occ off wh arg mtz, mnrf pyr nodes, rr qtz ovghs, mnrf slty lams, occ mic flks, lse, fri-mod hd, fr inf por, pr-fr vis por, fluor.

FLUOR: 6110°-6260°; 40-60% of SST, mod bri grn yel sptd-ptchy fluor, slw strmg cut, thk-thn flm res.

SILTSTONE: med-lt brn, med brn gry, com arg, occ sil, micmic i/p, hd-occ v hd, sbbiky, mnrf blkly-sbfiss.

SANDSTONE: lt trnsi brn, clr-trnsi i/p, f-med, rr crs, mod srt, sbang, occ ang-sbrnd, mod sil cmt, occ off wh-lt brn gry arg mtz, occ mic flks, fri-mod hd, rr lse, pr vis por, pr-fr inf por, fluor.

SILTSTONE: med brn, occ lt-dk brn, med brn gry, arg, rr aren, micmic, hd, sbbiky, occ blkly.

SANDSTONE: lt trnsi brn, clr-trnsi i/p, f-med, rr crs, mod srt, sbang, occ ang-sbrnd, mod sil cmt, occ off wh-lt brn gry arg mtz, occ mic flks, fri-mod hd, rr lse, pr vis por, pr-fr inf por, fluor.

SURVEY @ 6190.74°MDRT
INC: 91.66° AZI: 51.01°G

SILTSTONE: lt-med gry, med brn-brn gry, arg-occ aren, tr carb spks, micmic, mod hd-hd, sbbiky-blky.

